

LD9 Variable / Method Flow

Final Documentation Package for the Bank Marketing Subscription Prediction Project

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This document explains how the main variables, datasets, and method outputs flowed from the beginning of the project to the final recommendation.

1. High-Level Flow

1. Raw bank marketing data was loaded and audited.
2. Target variable and feature types were inspected.
3. Train and test sets were created to separate model training from final evaluation.
4. A preprocessing pipeline handled numerical and categorical variables.
5. Baseline models created the first benchmark.
6. Random Forest and bagging experiments extended the baseline stage.
7. StackingClassifier models were built and compared using cross-validation.
8. Manual OOF stacking created meta-features manually.
9. Blending created a simpler meta-feature workflow based on one split.
10. Three-layer stacking created an additional meta-learning layer.
11. A final comparison table consolidated all important test results.
12. The MVD proof checklist and business recommendation were produced.

2. Notebook-by-Notebook Variable Flow

Notebook	Important variables	Main outputs created
Notebook 1	dataset overview, course coverage topics	coverage matrix, selected dataset, project plan
Notebook 2	model_data, X_train, X_test, y_train, y_test, preprocessor	trained_baselines, baseline_results
Notebook 3	single_tree_pipeline, random_forest_100_pipeline, tree_feature_importance, rf_n_estimators_curve	RF comparison results, OOB results, max_features experiment
Notebook 4	base_models, cv_strategy, stacking_lr_pipeline, stacking_rf_pipeline	CV results, test stacking results
Notebook 5	oof_train_predictions, test_predictions_by_fold, meta_train, meta_test	manual OOF stacking result, blending result
Notebook 6	level1 meta-features, level2 model inputs, gradient boosting model	three-layer stacking result, final model comparison
Notebook 7	final_model_comparison,	final recommendation and

	mvd_proof_checklist, recommendation_table	proof tables
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3. Important Variable Meanings

- X_train and X_test: feature sets used for training and final testing.
- y_train and y_test: target labels used for training and final testing.
- preprocessor: the transformation block that handles categorical encoding and numeric preparation.
- trained_baselines: dictionary storing baseline model pipelines.
- baseline_results: comparison table of the first benchmark models.
- oof_train_predictions: matrix that stores fair base-model predictions for each training row during manual stacking.
- test_predictions_by_fold: test-set predictions made by each fold-trained base model; later averaged to build meta-test features.
- meta-features: new columns made from model prediction probabilities.
- final_model_comparison: one table that summarizes the major final model performances.
- mvd_proof_checklist: the proof table that maps each assignment requirement to a notebook.

4. Method Flow Summary

- Baseline stage: establish the first reference performance.
- Bagging/Random Forest stage: reduce variance and improve stability.
- Automatic stacking stage: combine different model strengths through a meta-model.
- Manual stacking stage: learn the stacking logic by creating OOF predictions manually.
- Blending stage: test a simpler but weaker alternative to OOF stacking.
- Three-layer stacking stage: check whether extra complexity is worthwhile.
- Final proof stage: consolidate the work into a clear project conclusion.